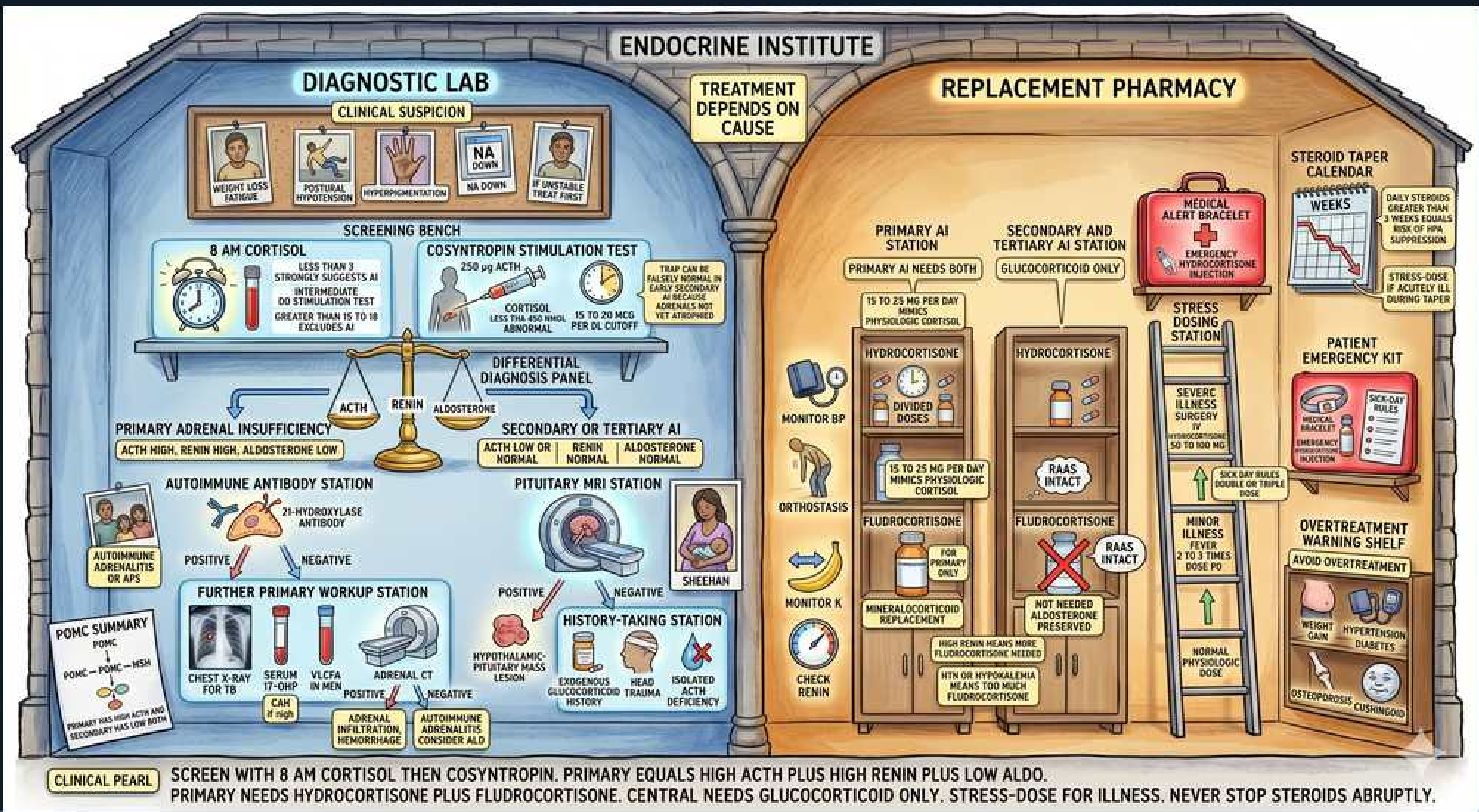
Adrenal Insufficiency/CAH — Diagnostic Algorithm Lab



1. Clinical suspicion board

WHAT YOU SEE

Top-left CLINICAL SUSPICION corkboard (weight loss, hyperpigmentation, low Na/high K)

WHAT IT ENCODES

Suspect AI in fatigue, weight loss, anorexia, hypotension/orthostasis, salt craving, hypoglycemia, and the lab clues hyponatremia + hyperkalemia. Hyperpigmentation points to PRIMARY AI specifically. Other supportive findings: eosinophilia, mild hypercalcemia, and a normocytic anemia. In acute decompensation (adrenal crisis), refractory shock not responding to fluids/pressors is the red flag.

BOARD TRAP

In suspected adrenal crisis, give stress-dose hydrocortisone immediately — do NOT delay treatment to complete diagnostic testing.

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2. 8 AM cortisol screen

WHAT YOU SEE

Alarm clock: 8 AM CORTISOL screening bench

WHAT IT ENCODES

Screening begins with a morning (8 AM) serum cortisol, drawn at the physiologic peak. A value <3 μg/dL is essentially diagnostic of AI; >18 μg/dL effectively excludes it. Intermediate values (3–18) are indeterminate and require a cosyntropin stimulation test. Interpret cortisol in light of binding proteins (estrogen/pregnancy and critical illness alter cortisol-binding globulin and total cortisol).

BOARD TRAP

A random afternoon/evening cortisol is uninterpretable for deficiency — only the 8 AM value is validated for screening.

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3. Indeterminate range

WHAT YOU SEE

Balance showing low vs intermediate vs adequate cortisol thresholds

WHAT IT ENCODES

Think of three buckets for the 8 AM cortisol: LOW (<3) confirms AI, HIGH (>18) excludes it, and INTERMEDIATE (3–18) is the gray zone that mandates dynamic testing. The cosyntropin stimulation test resolves the gray zone. This tiered approach avoids both over- and under-diagnosis from a single ambiguous level.

BOARD TRAP

An intermediate 8 AM cortisol is NOT reassuring — you must proceed to a cosyntropin stim test before excluding AI.

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4. Cosyntropin stim test

WHAT YOU SEE

COSYNTROPIN STIMULATION TEST station, check cortisol after ACTH

WHAT IT ENCODES

The standard high-dose cosyntropin (synthetic ACTH) test gives 250 μg IV/IM and measures cortisol at 30 and 60 minutes; a peak <18 μg/dL = AI. In primary AI the destroyed adrenal can't respond. In long-standing secondary/tertiary AI the atrophied adrenal also fails to respond. The test directly drives the adrenal, bypassing the hypothalamic-pituitary signal.

BOARD TRAP

In EARLY/acute secondary AI (e.g., recent pituitary surgery, fresh Sheehan/apoplexy) the not-yet-atrophied adrenal can give a falsely NORMAL response — use ACTH/insulin tolerance testing if clinical suspicion is high.

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5. Differential panel: ACTH

WHAT YOU SEE

DIFFERENTIAL DIAGNOSIS PANEL: high ACTH = primary; low/normal = secondary/tertiary

WHAT IT ENCODES

Once AI is confirmed, a simultaneous plasma ACTH localizes it: HIGH ACTH = primary (adrenal) AI; LOW or inappropriately-normal ACTH = secondary/tertiary (central) AI. This single value drives the rest of the workup (antibodies/CT for primary vs pituitary MRI for central). Pair ACTH with renin/aldosterone and DHEA-S for completeness.

BOARD TRAP

It is the ACTH level — not cortisol — that distinguishes primary from central AI; an 'inappropriately normal' ACTH with low cortisol still means central failure.

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6. Primary: high renin, low aldo

WHAT YOU SEE

PRIMARY AI label: ACTH high, renin high, aldosterone low

WHAT IT ENCODES

Primary AI loses the mineralocorticoid axis: aldosterone is LOW and renin is HIGH (the kidney is screaming for aldosterone it can't get). Combined with high ACTH and low cortisol/DHEA-S, this nails the adrenal as the failing organ. Clinically this is why primary AI has hyperkalemia, salt craving, and orthostasis.

BOARD TRAP

A high renin / low aldosterone pair confirms mineralocorticoid loss and therefore PRIMARY AI; these are normal in central AI.

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7. Secondary/tertiary: aldo intact

WHAT YOU SEE

SECONDARY/TERTIARY: ACTH low/normal, aldosterone normal

WHAT IT ENCODES

Central AI (secondary = pituitary, tertiary = hypothalamic/steroid) shows low cortisol with low/normal ACTH but PRESERVED aldosterone and renin, because the renin-angiotensin system — not ACTH — runs the glomerulosa. Hence normal potassium, no salt craving, pale skin, and only dilutional (cortisol-driven, ADH-mediated) hyponatremia.

BOARD TRAP

No hyperkalemia in central AI because aldosterone is intact — its presence should redirect you to a primary cause.

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8. Autoimmune antibody station

WHAT YOU SEE

AUTOIMMUNE ANTIBODY STATION: 21-hydroxylase antibodies

WHAT IT ENCODES

For confirmed PRIMARY AI, test 21-hydroxylase autoantibodies; a positive result confirms autoimmune adrenalitis (Addison disease), the #1 primary cause in the developed world. Positive patients should be screened for associated polyglandular autoimmunity (thyroid, T1DM = APS-2; candidiasis + hypoparathyroidism = APS-1).

BOARD TRAP

21-hydroxylase antibodies are the autoimmune marker; if they are NEGATIVE in primary AI, pursue TB, hemorrhage, metastasis, or adrenoleukodystrophy with imaging/VLCFA.

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9. Further primary workup

WHAT YOU SEE

FURTHER PRIMARY WORKUP: chest X-ray for TB, adrenal CT for hemorrhage/metastasis

WHAT IT ENCODES

Antibody-NEGATIVE primary AI needs structural and infectious evaluation: adrenal CT (bilateral enlargement/calcification suggests TB, infection, hemorrhage, or metastasis; small atrophic glands suggest autoimmune), chest imaging/testing for TB, and in young males very-long-chain fatty acids for X-linked adrenoleukodystrophy.

BOARD TRAP

Bilateral adrenal enlargement/calcification on CT points to TB, infiltration, hemorrhage, or metastasis — not autoimmune (which atrophies the glands).

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10. Pituitary MRI

WHAT YOU SEE

Pituitary MRI station for secondary AI

WHAT IT ENCODES

For LOW-ACTH (central) AI, image the pituitary/hypothalamus with MRI to find an adenoma, craniopharyngioma, metastasis, empty sella, Sheehan necrosis, apoplexy, or infiltration (hypophysitis, sarcoid). Also check the other pituitary axes (TSH/free T4, gonadotropins, IGF-1, prolactin) since deficiencies often coexist.

BOARD TRAP

Before replacing thyroid hormone in panhypopituitarism, replace glucocorticoid first — starting levothyroxine alone can trigger adrenal crisis.

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11. History-taking station

WHAT YOU SEE

HISTORY-TAKING STATION: exogenous steroids, head trauma, isolated ACTH deficiency

WHAT IT ENCODES

Always take a drug and exposure history: chronic exogenous glucocorticoids (oral, inhaled, topical, intra-articular) cause tertiary AI and are the most common cause of AI overall. Ask about head trauma, cranial radiation, postpartum hemorrhage (Sheehan), checkpoint-inhibitor immunotherapy, and steroidogenesis-blocking drugs (ketoconazole, etomidate, megestrol).

BOARD TRAP

A history of chronic steroid use (even inhaled/topical or megestrol) explains most central AI — missing it leads to missed adrenal crisis on withdrawal/stress.

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12. POMC summary

WHAT YOU SEE

POMC SUMMARY box bottom-left

WHAT IT ENCODES

POMC is the precursor the pituitary cleaves into ACTH and MSH. In primary AI, lost cortisol feedback drives POMC/ACTH up, raising MSH → hyperpigmentation. In central AI, POMC/ACTH are low → pale skin. This single mechanism links the lab finding (ACTH high vs low) to the exam finding (pigmented vs pale).

BOARD TRAP

Hyperpigmentation requires high ACTH/MSH — its presence rules IN primary AI, its absence with low ACTH supports central AI.

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13. Clinical pearl banner

WHAT YOU SEE

Bottom pearl: screen 8 AM cortisol then cosyntropin; primary = high ACTH/high renin/low aldo

WHAT IT ENCODES

Algorithm in one line: 8 AM cortisol (<3 confirms, >18 excludes) → cosyntropin stim (peak <18 = AI) → ACTH to localize (high = primary, low/normal = central) → renin/aldosterone/DHEA-S, then 21-hydroxylase antibodies/CT for primary or pituitary MRI for central. Treatment depends on cause: primary needs hydrocortisone PLUS fludrocortisone; central needs glucocorticoid only.

BOARD TRAP

Never withhold/abruptly stop replacement, and always stress-dose for illness or surgery to prevent crisis.

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14. Replacement pharmacy: hydrocortisone + fludrocortisone

WHAT YOU SEE

REPLACEMENT PHARMACY shelves: glucocorticoid for all, fludrocortisone added only for primary

WHAT IT ENCODES

Maintenance therapy: glucocorticoid replacement (hydrocortisone, usually split AM-larger/PM-smaller to mimic the diurnal rhythm) is given in ALL types of AI. Mineralocorticoid replacement (fludrocortisone) is added ONLY in primary AI, because central AI keeps aldosterone. Titrate hydrocortisone by clinical wellbeing/weight and fludrocortisone by blood pressure, orthostasis, sodium, potassium, and renin.

BOARD TRAP

Fludrocortisone is needed only in PRIMARY AI — adding it in secondary/tertiary AI is unnecessary because aldosterone is already intact.

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15. Stress-dosing station

WHAT YOU SEE

STRESS DOSING station / sick-day rules with severe illness escalation

WHAT IT ENCODES

Sick-day rules: double or triple the oral glucocorticoid for fever/moderate illness, and switch to parenteral stress-dose hydrocortisone (e.g., 100 mg IV then 50 mg q6–8h or infusion) for major stress, vomiting, surgery, or trauma. In a crisis, give high-dose hydrocortisone plus aggressive IV normal saline and dextrose; at stress doses hydrocortisone provides enough mineralocorticoid effect that fludrocortisone can be paused.

BOARD TRAP

At high stress doses of hydrocortisone, separate fludrocortisone is unnecessary (the glucocorticoid covers mineralocorticoid needs); never under-dose during intercurrent illness.

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16. Emergency kit + medical alert + taper calendar

WHAT YOU SEE

PATIENT EMERGENCY KIT, MEDICAL ALERT BRACELET, and STEROID TAPER CALENDAR on the right wall

WHAT IT ENCODES

Every AI patient needs an emergency injectable hydrocortisone kit, education on sick-day rules, and a medical-alert bracelet/card identifying steroid dependence. Patients on chronic steroids must be tapered slowly (the taper calendar) to let the suppressed HPA axis recover — never stopped abruptly. These outpatient measures are what prevent fatal adrenal crisis.

BOARD TRAP

Abrupt discontinuation of chronic steroids (skipping the taper) is a classic precipitant of adrenal crisis — every steroid-dependent patient needs alert ID, an emergency kit, and a taper plan.
